

ABHIRAMI THIAGARAJAN

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EDUCATION

The University of Texas at Austin Bachelor of Science, Computer Science May 2028
GPA: 4.0 /4.0

Relevant Coursework Data Structures and Algorithms, Introduction to Computing (ECE + BME),
Linear Algebra, Differential Equations

EXPERIENCE

AllWay – Co-Founder & AI Engineer; Remote Jan 2026 – Present

- Built agentic AI pipeline in Python connecting PostgreSQL, HuggingFace transformers, and Anthropic Claude API to answer natural language queries across 5 enterprise system classes with full source citation
- Designed three-layer retrieval architecture — structured SQL, vector similarity search, and domain knowledge graph — enabling cross-system operational intelligence across PLM, ERP, MES, QMS, and LIMS
- Optimised multi-step LLM pipeline using MPS GPU acceleration, asyncio parallel retrieval, and embedding caching — cut query latency from 120s to under 30s

University of Texas at Dallas – Research Intern; Dallas, TX Aug 2025 – May 2026

- Designed and shipped a modular FastAPI service handling multi-modal medical imaging requests — built validation middleware, async request queuing, and a pluggable preprocessing layer supporting 3 scan types.
- Profiled and resolved concurrency bottlenecks in a distributed processing pipeline — restructured worker scheduling and batching logic, increasing throughput by 40% across concurrent queues.
- Built CLI automation tools that replaced manual pipeline execution steps, reducing operator intervention and improving end-to-end pipeline reliability across distributed environments.

360DigiTMG – Software Development Intern; Dallas, TX May 2025 – Aug 2025

- Engineered Python/SQL ETL pipelines ingesting 50K+ daily records, cutting data latency to near real-time.
- Built and deployed a production Flask REST API with versioned prediction endpoints, request validation, and automated regression benchmarking on holdout data — served live traffic.
- Built a self-serve internal dashboard platform, eliminating 15+ hrs/week of ad-hoc analyst requests - reduced eng interrupt load

Wolfram – Software Research Intern; Boston, MA May 2024 – August 2024

- Engineered distributed backend across 250K+ nodes- refactored sequential jobs into parallelized pipelines, cutting runtime by 60%.
- Profiled and resolved memory and thread contention bottlenecks in GPU-accelerated molecular simulation pipelines — reduced peak memory usage by 35% on high-node workloads through targeted allocation and scheduling changes.

PROJECTS

CardioPulse – AI Cardiovascular Diagnostics Platform · Deployed March 2026

- Built and deployed a full-stack cardiovascular diagnostics platform — FastAPI backend serving real-time ECG analysis and cardiac risk prediction models, with a Plotly/Dash frontend validated with 5 clinical users.
- Deployed serverless inference on AWS Lambda + API Gateway — sub-second ECG response times for real-time clinical feedback.

CarbonAI – Sustainability Optimization Platform · Deployed February 2026

- Built and deployed a full-stack sustainability platform — serverless Python backend and Dash frontend delivering real-time AI-generated carbon reduction recommendations based on user-submitted emissions data.
- Architected stateless microservice APIs for LLM inference, carbon scoring, and recommendations across concurrent user sessions

ACCOMPLISHMENTS

- WWT Hackathon (1st Place, ~200 teams) – built CNN-based diagnostic imaging classifier 2026
- EnviHacks - Sustainability Innovation Award + Girls Recode Hacks (2nd Place) 2026
- Wolfram Emerging Leaders/Scholarship Award – Top 10% of research papers 2025
- General Electric STEAM Lead Award - instructed 100+ students in Python, engineering, robotics 2025

SKILLS

Technical Skills: Java, C++, Python, JavaScript, SQL, REST APIs, Git, AWS (Lambda, API Gateway, S3), HTML/CSS, LLM APIs

Certifications/Data Tools: AWS CCP, Azure Fundamentals, TensorFlow, Scikit-learn, Power BI

Organizations: Kappa Theta Pi (Pre-Professional Fraternity), Women in CS, Women in Engineering